

CARLETON UNIVERSITY

SCHOOL OF MATHEMATICS AND STATISTICS

STAT 1500 A

Introduction to Statistical Computing
Course Outline

Summer 2026

Instructor: Esam Mahdi

- E-mail: esammahdi@cunet.carleton.ca
- Office: 5240 Herzberg Laboratories (by appointment)
- Course website: <https://brightspace.carleton.ca/d2l/home/420371>
- Office hours: By appointment.

Lectures: Every Thursday from 5:35 pm to 8:35 pm (online via Zoom; the link is available in Brightspace announcements).

Important Dates:

- First day of classes: May 6, 2026
- Last day of classes: August 14, 2026
- No classes:
 - May 18, 2026
 - June 19 – 20, 2026
 - July 1, 2026
 - August 3, 2026

Academic Calendar: <https://calendar.carleton.ca/academicyear/>

Course Description: Basics of programming in R and introduction to statistical software; generating statistical plots; computing descriptive statistics; performing basic statistical procedures; fundamentals of numerical analysis; optimization; generating random numbers, performing simple simulations and simulation-based inference.

Course Materials The course will include a combination of instructor-prepared materials and curated external resources, including selected YouTube lectures.

All external materials will be clearly linked and aligned with course objectives. Students are expected to engage with these resources as part of their weekly learning activities.

Software: Statistical programming language R will be used.

Reference: R for Beginners, by Emmanuel Paradis, 2005

[YouTube Lectures: Sample1](#), [YouTube Lectures: Sample2](#)

Lab: Beginning on May 12

- A1: Tuesday 4:05 pm – 4:55 pm (online via Zoom; the link is available in Brightspace announcements by May 10)

- A2: Tuesday 5:05 pm – 5:55 pm (online via Zoom; the link is available in Brightspace announcements by May 10)

Evaluation: Your final grade for the course will consist of

1. Quizzes: **20%**
2. Assignments: **40%**
3. Project: **40%**

Note: The Project is further divided into **10% for the oral presentation** and **30% for the written report**.

Quizzes: Five quizzes (worth a total of 20% of your final grade) will be posted and submitted through Brightspace (navigate to Tools → Quizzes to access them). Late submissions will not be accepted, and no make-up work will be provided. Students will solve each quiz online and have only one attempt.

Table 1: Tentative schedule for Quizzes

Quiz	Percentage	Date to be assigned	Due date
Quiz 1	4%	Saturday, May 30 at 12:00 pm	Saturday, May 30 at 11:59 pm
Quiz 2	4%	Saturday, June 13 at 12:00 pm	Saturday, June 13 at 11:59 pm
Quiz 3	4%	Saturday, July 4 at 12:00 pm	Saturday, July 4 at 11:59 pm
Quiz 4	4%	Saturday, July 25 at 12:00 pm	Saturday, July 25 at 11:59 pm
Quiz 5	4%	Saturday, August 8 at 12:00 pm	Saturday, August 8 at 11:59 pm

Assignments: Four assignments (worth 40% of your final grade) will be posted and submitted through Brightspace (navigate to Tools → Assignments to access them). Late submissions will not be accepted, and no make-up work will be provided.

Table 2: Tentative schedule for assignments

Assignment	Percentage	Date to be assigned	Due date
Assignment 1	10%	Friday, May 8	Friday, May 29
Assignment 2	10%	Friday, May 29	Friday, June 26
Assignment 3	10%	Friday, June 26	Friday, July 17
Assignment 4	10%	Friday, July 17	Friday, August 7

Final Project: The final research project is a **group project (4-5 students per team)** and is worth **40% of your final grade** (10% for the oral presentation and 30% for the written report).

Team Formation: By the last week of June, the Instructor will assign project teams. Alternatively, students may choose their own team members. Once teams are finalized, the team member names will be announced on Brightspace.

Oral Presentation (10%): The oral presentation will be conducted online via Zoom during the last lecture on **August 13, from 5:35 pm to 8:35 pm**. Each team will have a total of **15 minutes: 10 minutes** for the presentation using PowerPoint (or similar slides), followed

by **5 minutes** for questions from the instructor and TA. The specific presentation time slot for each team will be posted on Brightspace, and the Zoom link will be provided on the day of the presentation. Each student will be evaluated individually using a rubric (see below).

Written Report (30%): The written report must be submitted as a **PDF file generated from an .Rmd file**. The report is due **August 13 at 11:59 PM**. Each student must submit the report individually via Brightspace using the same submission process as assignments (**Tools → Assignments**). **The report must clearly list the names and student IDs of all team members on the first page.** All submissions should be identical across team members.

Project Information: More detailed information about the project (including topic selection, data requirements, and rubrics for both the oral presentation and written report) will be announced on Brightspace in early July.

Late submissions for the written report will be penalized 50% per day (including weekends). For example, a report submitted on August 14 at 11:59 PM will receive a maximum grade of 50% of the earned mark; on August 15 at 11:59 PM, a maximum of 0%.

Each written report should include the following sections:

- **Title:** A clear and descriptive title for your project.
- **Team Information:** Names and student IDs of all team members.
- **Introduction:** Provide a literature review about the topic so that someone unfamiliar with it will be prepared to understand the rest of your project.
- **Methods and Output Results:**
 - Describe the methods you used to perform the analysis from beginning to end that you used to answer the research question.
 - Interpret and explain the results you obtain.
 - Ensure the tables and/or figures are labeled with table and/or figure numbers.
- **Discussion and Conclusion:**
 - Summarize what you found and discuss the implications or impact of your findings.
 - Discuss whether your results agree with what you expected to find.
 - Discuss whether and how your final model could be improved.
 - Discuss future questions or research that your study could lead to.
- **References:** Include all citations that you used in your project. The citation format is your choice – just be consistent. Make sure that you also include the source of the data set that you used in your study.

Oral Presentation Rubric (for individual evaluation): Each student will be evaluated individually using the rubric below. Each criterion is scored on a 0-10 scale.

Written Report Rubric (for team evaluation): The written report will be evaluated according to the following criteria. Each criterion is scored on a 0-10 scale, for a total of 40 points (which will be scaled to 30% of the final grade).

Table 3: Oral Presentation Grading Rubric (Individual)

Criteria	Very Good/Excellent (9-10)	Satisfactory/Good (6-9)	Needs Improvement (3-6)	Unsatisfactory (0-3)	Score
Organization & Structure	Presentation is well-organized. Ideas flow logically.	Presentation is adequately organized, but some sections could be better structured.	Presentation lacks clear organization. Ideas are disconnected.	Presentation is extremely disorganized and difficult to follow.	__/10
Visual Aids	Well-designed and add value to the presentation. They effectively support key points.	Visual aids are present but may not consistently enhance the presentation. They somewhat support key points.	Visual aids are present but do not add significant value. They minimally support key points.	Visual aids are absent or confusing.	__/10
Time Management	Presentation is within the time limit and follows the format precisely.	Presentation is slightly over/under the time limit and somewhat follows the format.	Presentation significantly exceeds or falls short of the time limit. Deviates noticeably from the format.	Presentation is excessively long/short and disregards the prescribed format.	__/10
Q&A Handling	Handles audience questions well. Provides thorough, well-reasoned responses.	Responds to audience questions adequately, but some answers lack depth or accuracy.	Struggles to address audience questions. Answers are often unclear or incorrect.	Unable to respond to audience questions satisfactorily.	__/10
Overall Impression	The presentation is very good and demonstrates a high level of competence.	The presentation is generally satisfactory, meeting the basic requirements.	The presentation is below average and lacks polish.	The presentation is extremely weak and fails to meet expectations.	__/10
Total (out of 50)					__/50

Note: For the oral presentation, each student is evaluated individually; all team members are expected to participate.

Policy for Missed Quizzes and Assignments:

1. **Missed Quizzes/Assignment:** Students are permitted to miss a maximum of one quiz and a maximum one assignment due to extenuating circumstances as defined by the [Academic Consideration Policy](#). The student must complete the [Academic Consideration for Coursework Request Form](#) within three business days of the original quiz/Assignment. Students who correctly follow this procedure will have the missed quiz/Assignment grade replaced by their grade on the final project. Failure to follow this procedure will result in a grade of 0% on the missed quiz/Assignment. Students who miss a second quiz/Assignment will be required to meet with the instructor to determine whether they will be permitted to remain in the course.
2. **Grading Concerns:** Concerns about grading on quizzes, assignments, or the project must be brought to the instructor's attention within 72 hours of these assessments being available for review. To avoid biased evaluation, you may only request a complete regrade of an entire quiz, assignment, or project. You may not request a regrade until you have reviewed the solutions and grading scheme for the assessment under consideration.

Table 4: Written Report Grading Rubric (Team)

Criteria	Excellent (9-10)	Good (7-8)	Satisfactory (5-6)	Needs Improvement (0-4)	Score
Problem Understanding & Significance	Clearly understands and articulates the problem's importance, scope, and relevance.	Understands the problem and its significance but may lack some depth or clarity.	Shows basic understanding of the problem's significance but lacks clear articulation.	Demonstrates limited understanding or fails to grasp the problem's significance.	__/10
Analysis & Interpretation	Analyzes data with depth, applies advanced techniques, and derives meaningful insights.	Conducts sound data analysis and provides reasonable interpretations.	Performs basic data analysis but lacks in-depth insights.	Provides limited, incorrect, or inadequate analysis and interpretations.	__/10
Documentation & Report Quality	Provides thorough, well-organized documentation with clear explanations and references.	Offers organized documentation with clear explanations.	Provides basic documentation but lacks organization or clarity.	Offers minimal, unclear, disorganized, or missing documentation.	__/10
Creativity & Innovation	Demonstrates exceptional creativity and introduces innovative elements to the project.	Shows creativity and incorporates some innovative aspects.	Displays basic creativity but lacks significant innovation.	Lacks creativity/innovation or fails to demonstrate any.	__/10
Total (out of 40)					__/40

Academic Integrity Policy: Misconduct in scholarly activity will not be tolerated and will result in consequences as outlined in Carleton University's Academic Integrity Policy (see <https://carleton.ca/registrar/academic-integrity/>). A list of standard sanctions in the Faculty of Science can be found here. Additional details about this process can be found on the Faculty of Science Academic Integrity website (<https://science.carleton.ca/academic-integrity/>). Students are expected to familiarize themselves with and abide by Carleton University's Academic Integrity Policy.

- **Plagiarism:** The University Academic Integrity Policy defines plagiarism as “presenting, whether intentionally or not, the ideas, expression of ideas or work of others as one’s own.” This includes reproducing or paraphrasing portions of someone else’s published or unpublished material, regardless of the source, and presenting these as one’s own without proper citation or reference to the original source. Examples of sources from which the ideas, expressions of ideas or works of others may be drawn from include but are not limited to: books, articles, papers, literary compositions and phrases, performance compositions, chemical compounds, art works, laboratory reports, research results, calculations and the results of calculations, diagrams, constructions, computer reports, computer code/software, material on the internet and/or conversations.
- Examples of plagiarism include, but are not limited to:
 - any submission prepared in whole or in part, by someone else, including the unauthorized use of generative AI tools (e.g., ChatGPT);
 - using ideas or direct, verbatim quotations, paraphrased material, algorithms, formulae, scientific or mathematical concepts, or ideas without appropriate acknowledgment

- in any academic assignment;
- using another’s data or research findings without appropriate acknowledgement;
- submitting a computer program developed in whole or in part by someone else, with or without modifications, as one’s own; and
- failing to acknowledge sources through the use of proper citations when using another’s work and/or failing to use quotations marks.
- Plagiarism is a serious offence that cannot be resolved directly by the course’s instructor.

The Associate Dean of the Faculty follows a rigorous [process for academic integrity allegations](#), including reviewing documents and interviewing the student, when an instructor suspects a violation has been committed. Penalties for violations may include a final grade of “F” for the course.

Academic Accommodation

You may need special arrangements to meet your academic obligations during the term. For an accommodation request the processes are as follows:

Pregnancy obligation: write to me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details see the [Student Guide](#).

Religious obligation: write to me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details see the [Student Guide](#).

Students with Disabilities: The Paul Menton Centre for Students with Disabilities (PMC) provides services to students with Learning Disabilities (LD), psychiatric/mental health disabilities, Attention Deficit Hyperactivity Disorder (ADHD), Autism Spectrum Disorders (ASD), chronic medical conditions, and impairments in mobility, hearing, and vision. If you have a disability requiring academic accommodations in this course, please contact PMC at 613- 520-6608 or pmc@carleton.ca for a formal evaluation. If you are already registered with the PMC, contact your PMC coordinator to send me your Letter of Accommodation at the beginning of the term, and no later than two weeks before the first in-class scheduled test or exam requiring accommodation (if applicable). After requesting accommodation from PMC, meet with me to ensure accommodation arrangements are made. Please consult the PMC website (<http://carleton.ca/pmc>) for the deadline to request accommodations for the formally-scheduled exam (if applicable).

★ The course outline is subject to change based on the dynamics of the course ★